File 28: Intern Workstreams & Operational Training Playbook

> ***B2B Internship Operation Manual & Technology Blueprint**: - **Unified Infrastructure Stack**: All operations at Blue Ridge Stream are centralized across five standard software platforms: **AWS** (S3 data buckets, EC2 spatial compute, and RDS databases), **GitHub** (version control for CAD models and code pipelines), **QuickBooks Online** (financial accounting and margin carry reconciliations), **EspoCRM** (the premier open-source Salesforce alternative, deployed via Docker on AWS EC2), and **Google Drive** (central document repositories and GIS shapefiles). - **12-Page Intern Field Playbook**: This document defines three specialized, 4-page operational workstreams designed to delegate active project construction, biological permitting, and B2B sourcing pipelines to incoming university interns.*

I. Workstream 1: Fluvial Geomorphology & CAD/GIS Engineering Intern

- **Primary Position Focus**: Fluvial Geomorphology, AutoCAD Civil 3D geomorphic surface modeling, HEC-RAS 2D flow simulations, and USACE Savannah SOP credit yield calculations.
- **University Profile**: Georgia Tech / UGA Fluvial Engineering (Advanced Civil/Hydrology majors).

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AutoCAD Civil 3D Surface Modeling & HEC-RAS 2D Hydraulic Flow Simulations

1. Geomorphic Design Construction Layout
Interns are responsible for importing high-density field total-station and RTK GPS survey data into AutoCAD Civil 3D to construct the existing digital terrain model (DTM). The intern must then execute the following design adjustments:

- **Narrow the Width-to-Depth Ratio (W_{bkf}/d_{bkf})**: Using NCD methods, design a narrow bank-full channel width to accelerate water velocity during low-flow periods, keeping sediment moving and scouring deep, cold trout holding pools.
- **Instream Log Cross-Vanes & Boulder J-Hooks**: Overlay instream wood and rock structures. J-Hooks must be angled at 20 to 30 degrees to the bank, deflecting high-velocity stream flows toward the center of the channel to protect the bank from erosion.
- **High-Gradient Bank Regrading**: Design stable bank slopes ($\le 3:1$ or $4:1$) integrating native soil lifts and geomechanical geogrids to bind unstable bank soils.

2. HEC-RAS 2D Hydrodynamic Modeling
To verify that the geomorphic designs will survive peak storm events, the intern must compile and run a HEC-RAS 2D model:

- **Import the Proposed Civil 3D XML Surface**: Export the design corridor surface as LandXML and import it as the terrain layer in RAS Mapper.
- **Generate 2D Mesh & Boundary Conditions**: Set a 2-foot mesh cell spacing. Set boundary condition lines for the upstream inlet (simulating 10-year and 100-year peak-flow storm events using USGS regression equations) and downstream outlet (normal depth).

- **Run Hydraulic Stress Simulations**: Calculate instream velocity (V), water surface elevation (WSE), and boundary shear stress (*au*). Verify that the geomorphic design keeps shear stress below the failure limit of the geogrids, ensuring a safety factor $SF_{shear} \ge 1.5$.

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Rosgen Level III Fluvial Hydrology & Sediment Transport Equations

1. Fluvial Hydrology & BANCS Model Audit
Interns must audit the existing geomorphic variables using the Rosgen Level III framework to calculate sediment supply and stream bank erosion rates:

- **Bank Erosion Hazard Index (BEHI)**: Physically measure bank height, root depth, root density, bank angle, and surface protection to assign a BEHI score (Low, Moderate, High, Extreme) for each reach.
- **Near-Bank Shear Stress (NBS)**: Calculate NBS by measuring the ratio of near-bank maximum depth to mean channel depth (d_{nb}/d_{mean}) or using velocity profile distributions.
- **Erosion Estimation**: Plot the BEHI and NBS variables on the Savannah District validation curves to estimate stream bank erosion rates in cubic yards per linear foot per year.

2. Sediment Transport & Spawning Bed Flushing Mechanics
To prevent siltation of wild trout spawning gravels, interns must apply the dual-Shields bed flushing equations:

- **Determine Critical Shear Stress (*au_c*)**: Calculate the shear stress required to move the median bed material ($d_{50} = 15\text{--}40\text{ mm}$):

$$\tau_c = \theta_c \cdot (\gamma_s - \gamma_w) \cdot d_{50}$$

where $\theta_c = 0.045$ is the dimensionless Shields parameter, γ_s is sediment unit weight, and γ_w is water unit weight.

- **Flushing Stress Window**: Verify that under bank-full velocity, the channel shear stress satisfies the flushing window ($1.46\text{ N/m}^2 < \tau < 21.85\text{ N/m}^2$). This ensures that high water flows self-clean fine clay silt without moving the heavy trout spawning gravels.

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AWS Spatial Compute Clusters & GitHub Version-Controlled CAD DWG Pipelines

1. AWS EC2 Spatial Compute Cluster Setup
Large-scale AutoCAD Civil 3D rendering and HEC-RAS 2D hydraulic simulations require intensive CPU compute. Interns must deploy spatial compute nodes on AWS:

- **Spin Up AWS EC2 Compute Node**: Deploys an AWS EC2 instance (e.g. `c6i.4xlarge` featuring 16 vCPUs and 32GB RAM) running Ubuntu Server.
- **Install Geospatial Software Dependencies**: Install the GDAL spatial libraries and compile the USACE HEC-RAS engine locally:

```
  sudo apt-get update && sudo apt-get install -y gdal-bin libgdal-dev build-essential gfortran
  # Sync terrain and bathymetry data from the central AWS S3 spatial bucket
  aws s3 sync s3://blueridgestream-spatial-data/terrain/ ./local_terrain/
```

- **Execute Batch Hydraulic Simulations**: Execute HEC-RAS batch command simulations via the AWS CLI, outputting high-fidelity flood inundation spatial datasets.

2. GitHub Version-Controlled CAD Pipelines

Because CAD binary drawings (`.dwg` or `.dxf` formats) are large and cannot be natively compared using line-by-line Git diffs, interns must deploy a version-controlled CAD pipeline on GitHub:

- **Setup Git LFS (Large File Storage)**: Initialize Git LFS in the project GitHub repository to handle binary CAD files:

```
git lfs install
git lfs track "*.dwg"
git lfs track "*.dxf"
git add .gitattributes
```

- **Version-Control & Pull Request Workflows**: Establish a feature-branch system. Every geomorphic design iteration must be committed with a detailed Markdown commit message describing the geomorphic slope changes. Merge changes into the `main` branch only after a peer review by Hunter Morris.

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Stream SOP Credit Yield Calculations & USACE Savannah District SOP Verification Protocols

1. Savannah SOP Credit Yield Math
Interns must calculate the net credit yield of proposed projects using the USACE Savannah District Standard Operating Procedure (SOP) formulas:

- **Formula for Net Credit Lift (CL_{net})**:

$$CL_{net} = (M_{post} - M_{pre}) \cdot L_{reach} \cdot S_{factor}$$

where M_{pre} is the pre-restoration mitigation matrix score (typically 1.2 to 2.2), M_{post} is the post-restoration design score (targeting 6.8 to 8.6 under Priority 1 NCD rules), L_{reach} is reach length, and S_{factor} is the watershed significance factor.

- **Compute Project Yield**: For Roya's Cabin Anderson Creek case study ($L_{reach} = 1{,}500\text{ LF}$): Calculate the geomorphic lift ($M_{post} - M_{pre} = 8.6 - 2.2 = 6.4$) to yield **12,900 stream credits.**

2. USACE Savannah District IRT Verification Package Compilation
The intern must compile the final Mitigation Banking Instrument (MBI) coordinate documents for the Interagency Review Team (IRT) submission:

- **GIS Shapefiles**: Export AutoCAD boundary lines as ESRI Shapefiles, projecting them in UTM Zone 17N (NAD83 Georgia State Plane).
- **savannah District SOP Worksheet**: Complete the Excel SOP worksheets. Ensure every field is fully calculated with zero placeholders, including the Soil Erodibility Factor (K-value) and drainage basin square mileage.
- **USACE MBI Package Checklist**: Compile the 2D HEC-RAS flood maps, the Rosgen Level III geomorphic BANCS charts, and the water temperature monitoring logs into the final USACE Savannah District submittal package on Google Drive.

II. Workstream 2: Coldwater Ecology & Environmental Permitting Intern

- **Primary Position Focus**: Coldwater fisheries biology, wild trout habitat spawning restoration, thermal budget shading analysis, macroinvertebrate sampling, and Georgia EPD stream buffer variance permitting.
- **University Profile**: UGA Odum School of Ecology / Clemson Environmental Sciences (Biology majors).

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Brook Trout Spawning Gravel Geomorphic Specs & Native Rhododendron Buffer Shade Formula Audits

1. Spawning Gravel Geomorphic Specifications

To ensure the successful re-colonization of wild Brook, Rainbow, and Brown trout, interns must physically measure and sort the spawning gravel substrates:

- **Median Substrate Size (d_{50})**: Substrate samples must fall within the range of **15 to 40 mm** (clean, rounded mountain river gravels).
- **Fine Sediment Constraint**: Sieve analysis must demonstrate that fine clays and sands ($<2\text{ mm}$) represent less than **8 percent** of the total substrate volume, preventing the smothering of trout eggs in spawning nests.
- **Instream Placement**: Place sorted spawning gravels directly at pool tailouts where upwelling water provides oxygen to the eggs.

2. Thermal Cooling Shade Budget & Solar Attenuation Formulas

Interns must audit the shading canopy design to ensure stream summer water temperatures stay below the critical **65 degrees F** wild trout spawning limit:

- **Solar Radiation Attenuation Model**: Calculate the solar heat load reduction (Q_{net}) under the mature native Rhododendron maximum buffer canopy:

$$Q_{net} = Q_{incoming} \cdot e^{-\kappa \cdot \text{LAI}}$$

where $Q_{incoming}$ is open solar radiation, $\kappa = 0.65$ is the light extinction coefficient, and $\text{LAI} = 5.2$ is the leaf area index of the Rhododendron canopy.

- **Verify Attenuation**: Calculate that the mature canopy blocks **96 percent** of incoming solar radiation, maintaining the coldwater trout spawning thermal window.

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Macroinvertebrate EPT Index Bio-Assessments & Stream Water Quality/Thermal Sampling Logs

1. Macroinvertebrate EPT Index Bio-Assessments

To verify the biological recovery of the stream, interns must conduct regular macroinvertebrate sampling and calculate the Ephemeroptera, Plecoptera, and Trichoptera (EPT) index:

- **Field Sampling**: Collect samples from three riffle sections using a standard 500-micron D-frame kick net.
- **Laboratory Identification**: Identify specimens to the family level using a dissecting microscope. Count the total families belonging to the three critical clean-water orders:
- *Ephemeroptera (Mayflies)*: Baetidae, Heptageniidae.
- *Plecoptera (Stoneflies)*: Perlidae, Pteronarcyidae.
- *Trichoptera (Caddisflies)*: Hydropsychidae, Limnephilidae.
- **EPT Index Calculation**: Calculate the EPT score. A clean, healthy mountain stream must yield an EPT score **> 15**, verifying excellent water quality.

2. Water Quality & Thermal Logger Protocols

Interns must maintain the network of on-site water quality sensors and loggers:

- **Deploy HOBO Water Temp Loggers**: Secure HOBO loggers inside steel protective pipes anchored deep within pools. Program them to log water temperatures every 15 minutes.

- **Download & Calibrate Data**: Download data monthly using the HOBO Mobile app. Calibrate the loggers against a NIST-certified thermometer to ensure high data quality.
- **Water Quality Testing Logs**: Measure dissolved oxygen (DO must be *\ge 7.0\text{ mg/L}*), pH ($6.5\text{--}8.0*), and turbidity (*<10\text{ NTU}$) using a calibrated multi-parameter water quality probe, logging all measurements in the central Google Drive database.

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DNR EPD State Stream Buffer Variance Filing Protocols & NEPA Permitting Review Cycles

1. DNR EPD State Stream Buffer Variance Applications
Under Georgia law, any activity within the standard **25-foot state stream buffer** (or **50-foot trout stream buffer**) requires a variance from the Georgia DNR Environmental Protection Division (EPD). Interns must compile and file these applications:

- **Select the Buffer Variance Category**: File under *Category 1 (Restoration Projects)*, which allows buffer modifications that improve water quality and stream geomorphology.
- **Prepare the Buffer Mitigation Plan**: Draft a detailed revegetation plan incorporating only native streamside vegetation (such as mature native Rhododendron maximum, Mountain Laurel, and Black Willows) at a density of 400 plants per acre.
- **Compile the Engineering Exhibits**: Coordinate with the geomorphology intern to attach HEC-RAS 2D flood maps and AutoCAD drawings showing that the bioengineered restoration will not increase water velocity or erosion.

2. USACE Nationwide Permit 27 (NWP 27) & NEPA Compliance
Stream restorations are typically permitted under USACE Nationwide Permit 27 (Aquatic Habitat Restoration). Interns must track compliance with the National Environmental Policy Act (NEPA):

- **Complete the Pre-Construction Notification (PCN)**: Complete the USACE Form 4345. Attach the geomorphic designs, the biological EPT baselines, and the EPD buffer variance application.
- **Coordinate with Regulatory Agencies**: Coordinate with USACE, EPA, Fish and Wildlife Service, and DNR EPD during the 45-day agency review period. Keep track of all regulatory comments on the central project Google Drive.

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AWS S3 Database Syncing of Field Drone Telemetry & Thermal Mapping Spatial Datasets

1. AWS S3 Field Data Storage Architecture
To support the interactive canvas drone simulator on the portal, interns must organize and sync field drone data using AWS S3 storage buckets:

- **AWS S3 Bucket Structure**: Organize the `blueridgestream-field-telemetry` bucket into logical subdirectories for each project:

```
s3://blueridgestream-field-telemetry/projects/anderson-creek/orthomosaics/
s3://blueridgestream-field-telemetry/projects/anderson-creek/thermal-scans/
s3://blueridgestream-field-telemetry/projects/anderson-creek/hobo-temperature-logs/
```

- **Setup AWS S3 Lifecycle Policies**: Configure lifecycle policies to transition raw drone footage and heavy orthomosaics to AWS S3 Glacier Deep Archive after 90 days, reducing storage costs by 75 percent.

2. Programmatic Syncing via AWS CLI

Interns must run programmatic scripts to sync field datasets directly from on-site laptops to AWS S3:

- **Write the AWS S3 Upload Script**: Interns write a Python script that runs locally on field laptops to sync newly collected HOBO water temperature logs and drone thermal scans:

```
import boto3
import os

s3 = boto3.client('s3')
bucket_name = 'blueridgestream-field-telemetry'
local_folder = './field_data/anderson-creek/'

for filename in os.listdir(local_folder):
    if filename.endswith('.csv') or filename.endswith('.tif'):
        local_path = os.path.join(local_folder, filename)
        s3_path = f'projects/anderson-creek/{filename}'
        s3.upload_file(local_path, bucket_name, s3_path)
        print(f'Uploaded {filename} to AWS S3!')
```

- **Setup Google Drive Backup Integration**: Automate a daily sync between Google Drive folder updates and the AWS S3 raw backup bucket to keep the entire team aligned.

III. Workstream 3: B2B Business Development, Landowner Sourcing & CRM Intern

- **Primary Position Focus**: B2B sales development, landowner joint-venture sourcing, county records search, open-source CRM architecture, and project carry-cost financial reconciliations.
- **University Profile**: Clemson / UGA Terry College of Business (Real Estate/Finance/B2B Sales majors).

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Open-Source CRM (EspoCRM/SuiteCRM) Pipeline Architecture & Google Drive CRM Integrations

1. Open-Source CRM Deployment & Lead Pipeline Architecture

To avoid expensive Salesforce licensing fees, Blue Ridge Stream uses **EspoCRM**, the leading open-source CRM alternative. Interns are responsible for managing EspoCRM, deployed via Docker on an AWS EC2 instance with an AWS RDS MySQL database:

- **EspoCRM Custom Fields**: Set up custom fields for the stream business, including HUC-8 River Basin, Stream Length (Linear Feet), Estimated USACE Credit Yield, and Landowner JV Split Ratio.
- **Sales Pipeline Stage Mapping**: Interns track deals through five specific pipeline stages:

`1. Lead Sourcing (County Deed Audit)` *\rightarrow* `2. Outreach (Initial Call/Stream Walk)` *\rightarrow* `3. Geomorphic Valuation` *\rightarrow* `4. USACE IRT Permitting` *\rightarrow* `5. Signed JV (70/30 Contract)`

2. Automated Google Drive & CRM Integrations

To keep land deeds and survey maps organized, interns manage automated integrations between the CRM and Google Drive:

- **Setup Automated Folder Generation**: Using EspoCRM Webhooks connected to Google Drive, configure the CRM to automatically generate a standard Google Drive folder structure when a lead is moved to the 'Geomorphic Valuation' stage:

`Google Drive/ACR Stream Restoration/03_Project_Case_Studies/[Project_Name]/`

```
■■■ 01_AutoCAD_Designs/
■■■ 02_USACE_SOP_Permitting/
■■■ 03_Land_Deeds_&_Contracts/
■■■ 04_Field_Drone_&_Photos/
```

- **Maintain Google Drive Database Integrity**: Ensure all signed land option agreements and USACE SOP worksheets are saved in their respective project folders.

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County Record Deed Audits, Senior Mortgage Liens, & Escrow Covenants Sourcing Steps

1. County Tax Assessor Sourcing Steps
Interns use county GIS platforms (such as Qpublic) to identify properties in North Georgia (Fannin, Union, Gilmer, Lumpkin counties) with degraded streams:

- **Filter Properties**: Filter for agricultural, forestry, or recreational parcels with at least 50 acres and 1,000 linear feet of stream channel.
- **Analyze Erosion via Orthomosaics**: Inspect high-resolution aerial imagery to locate severe stream bank erosion, mud-laden cattle access points, or washed-out crossings.
- **Export Owner Metadata**: Extract the owner's legal name, parcel ID, mailing address, and deed book/page numbers, importing them directly into EspoCRM.

2. Legal Due Diligence: Title Audits & Mortgage Subordinations
Before Blue Ridge can invest capital into a stream restoration JV, interns must perform a title audit to manage financial risk:

- **Conduct Deed Registry Search**: Log into the Georgia Superior Court Clerks' Cooperative Authority (GSCCCA) database. Search the property's historical title chain to identify any outstanding mortgages, tax liens, or easements.
- **Manage Mortgage Lien Risks**: If the property is mortgaged, the land easement requires a recordable **Mortgage Subordination Agreement**. The intern must draft this agreement using the Blue Ridge legal resolution template, ensuring that the conservation easement survives foreclosure.
- **Establish Landowner Mitigation Escrow Account (LMEA)**: For mortgaged properties, structure an escrow account to capture a portion of credit sales, providing a backup to pay down the mortgage and secure the bank's approval.

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70/30 Landowner JV Split Pitching Playbooks & Bank Special Asset Relationship Management

1. Managing Bankers & Lawyers Referrals
Interns are responsible for managing relationships with the 50 relationship bankers (File 24) and 50 environmental/real estate lawyers (File 25) in Georgia:

- **Execute Sourcing Campaigns**: Send targeted email campaigns via EspoCRM, using the banker and lawyer outreach templates. Manage follow-ups and schedule virtual meetings.
- **Bank Special Assets Pitches**: Track bank foreclosures on Fannin and Union county land holdings. Pitch special asset managers on using a zero-CapEx stream JV to monetize non-performing land assets, helping the bank recover outstanding debt.

2. Sourcing Landowners with the 70/30 Trout Pitch

Interns use a structured B2B sales playbook to close JVs with private landowners, presenting a premium alternative to high-overhead corporate competitors:

- **The RES Flat-Rate Sourcing Counter-Play**: Educate landowners that RES and EIP offer flat-rate credit royalties ($12–$25 per credit) which capture all the upside. Under Blue Ridge's 70/30 split, the landowner collects 30 percent of the actual market price ($33/credit at $110/credit), generating **more than double** the corporate payout.
- **Aesthetic & Financial Value Comparison**: Present a custom project model demonstrating how a 1,500-foot stream bank JV (such as Anderson Creek) yields **425,700 USD** in cash proceeds for the landowner, while building a pristine Brook Trout fly-fishing creek at **0 USD cost** to the owner.

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QuickBooks Online Financial Reconciliation & Project Margin Carrying Cost Math

1. QuickBooks Online Accounting & Reconciliation

Interns manage the corporate books and track capital accounts in **QuickBooks Online**:

- **Invoice B2B Credit Sales**: When stream credits are sold (e.g. 5,000 credits sold to a data center client at $110/credit, totaling $550,000), generate the invoice in QuickBooks.
- **Reconcile JV Escrow Distributions**: Reconcile cash allocations according to the 70/30 contract. Transfer the landowner's 30 percent share ($165,000) and allocate the developer's 70 percent share ($385,000) to operational accounts.
- **Fund the Operational Escrows**: Transfer 12.5 percent of gross proceeds to the Working Capital Reserve Escrow (WCRE) account in QuickBooks, ensuring strict compliance with the bylaws.

2. Project Margin & Carry Cost Math

Interns run financial models to evaluate capital carry costs and investment return (ROI) metrics:

- **Calculate Developer Project Carrying Cost (*CC*)**:

$$CC = \text{CapEx} \cdot (1 + r)^t - \text{CapEx}$$

where $\text{CapEx} = 345,000\text{ USD}$, $r = 8.5\%$ is the annual bank construction interest rate, and $t = 1.25\text{ years}$ is the permitting/construction cycle.

- **Verify Margin Calculations**: Reconcile carry costs ($CC = 37,284\text{ USD}$) against developer net proceeds ($993,300\text{ USD}$) to ensure the project Net Profit Margin exceeds **40 percent**, verifying the financial health of the stream venture.